

Departement Elektriese, Elektroniese en
Rekenaar-Ingenieurswese

Department of Electrical, Electronic and
Computer Engineering

Eksamen

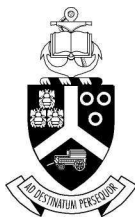
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Examination

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Module EBN111
Elektrisiteit en Elektronika
4 Junie 2010

Module EBN111
Electricity and Electronics
4 June 2010



UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

Eksamensinligting:

Exam information:

Maksimum punte: <i>Maximum marks:</i>	90	Volpunte: <i>Full marks:</i>	90
Duur van vraestel: <i>Duration of paper:</i>	180 minute <i>180 minutes</i>	Oopboek / toeboek: <i>Open / closed book:</i>	Toe <i>Closed</i>

Memo

Dosent(e): <i>Lecturer(s):</i>	1 J Joubert		
	2 D Bhatt		
	3 B Mabuza		
Groephoof: Ek bevestig dat die vraestel die uitkomstes toets soos gespesifiseer in die studiehandleiding. <i>Group head: I confirm that the question paper evaluates the outcomes as specified in the study guide.</i>			
Groephoof: <i>Group Head:</i>	Prof J Joubert	Handtekening: <i>Signature:</i>	

EBN111 EKSAMEN ANTWOORDBLAD/ EXAM ANSWER SHEET

4 JUNIE/JUNE 2010

Student se besonderhede/Student's details									
Van en voorletters: <i>Surname and initials:</i>					Studentenommer: <i>Student number:</i>				
Graadkursus: <i>Degree Course:</i>									
Tel. nr.: <i>Tel no.:</i>									

PUNTE/MARKS									
Q1	/30	Q2	/25	Q3	/15	Q4	/20	Final	/90

VRAAG /QUESTION 1 [30]		
1.1	$W = 270MJ$	$N_{bat} = 10$
1.2	$Q = 2160C$	$P = 0.292W$
1.3	$P_V = -5W$	$P_I = -40W$
1.4	$G_{eq} = 1S$	
1.5	$I_3 = 12mA$	$V_2 = 15V$
	$I_3 = 12mA$	
1.6	$a = 3$	$b = 10$
	$c = 6$	$d = 60$
1.7	$e = 2$	$f = 15$
	$g = 12$	$h = 50$
1.8	$\Delta = 14$	$v_2 = 1.43V$
1.9	$I_B = 40.6\mu A$	$V_{CE} = 12.475V$
1.10	$I = 7.5\mu A$	$v_0 = -15V$

VRAAG /QUESTION 2 [25]		
2.1	$v_o = 40V$	
2.2	$V_s = 50V$	$R_s = 300\Omega$
2.3	$v_{1(10V)} = -1.43V$	$i_{1(0.5A)} = -0.18A$
2.4	$R_L = 3.33K \Omega$	$P_{max} = 83.4mA$
2.5	$R_{TH} = 6.55 \Omega$	
2.6	$V_{TH} = 312V$	$R_{TH} = 72 \Omega$
2.7	$R_{TH} = 66.7 \Omega$	
2.8	$I_N = 7.5mA$	$R_N = 85.7 \Omega$
2.9	$R_N = 0 \Omega$	

VRAAG 3 – QUESTION 3 [15]			
3.1	$v_o = 8mV$		$i_o = -2\mu A$
3.2	$V_{out} = -15V$		
3.3	$V_1 = -60mV$	$V_2 = 0.5mV$	$V_3 = 602.75mV$
3.4	$L_{eq} = 11.5H$		
3.5	$C_{eq} = 35\mu F$		
3.6	$W_C = 400nJ$	$W_L = 0.18J$	$W_{10\Omega} = 5.76Kj$ OR $1.6Wh$

VRAAG 4 – QUESTION 4 [20]		
4.1	(a) $v(t) = 24.8 \cos (\omega t + 36^\circ) \text{ V}$	
	(b) $\bar{V}_1$, 160°	
	(c) $\bar{V}_2 = 0 + j2 \text{ V}$	
4.2	(a) $0.38 + j0.87$	
	(b) $3.76 \angle 50.8^\circ$	
4.3	$\bar{a} = 2 + j5$	$\bar{b} = -j5$
	$\bar{c} = -j5$	$\bar{d} = 1 + j3$
4.4	(a) $i(t) = 2.06 \cos (400t - 36.7^\circ) \text{ A}$	
	(b) $Z_{ab} = 1.47 \angle 79.4^\circ \Omega$	
4.5	$\bar{V}_{th} = 8 \angle 173.13^\circ \text{ V}$	